Uniform Random Mapping Functional-Graph Closure: Exact Cycle–Component Counts and Marked-Orbit Limits

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Abstract

For a uniform random function $f : [n] \to [n]$, we give a self-contained uniform random mapping functional-graph closure. Decomposing the graph into a permutation and a rooted forest yields the exact joint count of cyclic vertices and components, its marginal, and every expected cycle count. An ordered-prefix argument gives the complete tail–cycle law of a marked orbit, identifies its collision length in distribution with the number of cyclic vertices, and proves their common Rayleigh limit.

1 Model and theorem

Choose each of the n^n functions $f : [n] \to [n]$ with equal probability. Its directed functional graph has edges $v \to f(v)$. Let C_n be the number of cyclic vertices and K_n its number of weak components. Write $c(k, r)$ for the unsigned Stirling number of the first kind and $(n)_k = n(n-1)\cdots(n-k+1)$.

Theorem 1 (Finite and asymptotic closure)**.** *For $1 \le r \le k \le n$,*

$$\#\{f : C_n = k, K_n = r\} = \binom{n}{k} c(k, r)\, kn^{n-k-1}, \tag{1}$$

where the last factor means the unique empty forest when $k = n$. Hence

$$\mathbb{P}(C_n = k) = \frac{(n)_k\, k}{n^{k+1}}, \qquad \mathbb{E}Z_{n,\ell} = \frac{(n)_\ell}{\ell n^\ell}. \tag{2}$$

where $Z_{n,\ell}$ counts directed ℓ-cycles. Fix a starting vertex. If μ is its tail length, λ its eventual cycle length, and $R_n = \mu + \lambda$, then

$$\mathbb{P}(\mu = u, \lambda = \ell) = \frac{(n-1)_{u+\ell-1}}{n^{u+\ell}} \quad (u \ge 0,\ \ell \ge 1,\ u + \ell \le n). \tag{3}$$

$R_n \stackrel{d}{=} C_n$, and both variables divided by $\sqrt{n}$ converge to the Rayleigh density $xe^{-x^2/2}\mathbf{1}_{x\ge 0}$.

These distributions are classical random-mapping statistics [1, 2]. Our contribution is a source-locked proof synthesis and executable boundary audit, not a literature-priority claim.

2 Permutation and forest proof

Fix the set S of k cyclic labels. The restriction to S is a permutation, and $c(k, r)$ permutations have r cycles. Every label outside S lies in an in-tree directed toward a root in S. The corresponding complete-graph Laplacian minor, of size $m = n - k$, is

$$L_S = nI_m - J_m.$$

For $m > 0$ its eigenvalues are k once and n with multiplicity $m - 1$, so $\det L_S = kn^{n-k-1}$. At $m = 0$ the empty determinant is one. Choosing S, its permutation, and this forest is bijective with the maps in (1).

Summing over r and using $\sum_r c(k, r) = k!$ gives

$$\#\{f : C_n = k\} = \binom{n}{k} k!\, kn^{n-k-1}.$$

After division by n^n, this is the first formula in (2). There are $\binom{n}{\ell}(\ell-1)! = (n)_\ell/\ell$ possible directed ℓ-cycles. Each prescribes ℓ independent function values and has probability $n^{-\ell}$, so indicator linearity proves the second formula. In particular, $n = 1$ gives the single fixed-point map, while $k = n$ reduces (1) to $c(n, r)$ as it must for permutations.

3 Marked orbit and the Rayleigh bridge

Put $t = u + \ell$. Before its first repeat, the marked orbit visits t distinct labels. After the fixed starting label, the ordered prefix has $(n-1)_{t-1}$ choices. Its next edge is forced to the prefix label at index u, and all $n-t$ unexposed function values remain arbitrary. The cell therefore contains $(n-1)_{t-1}n^{n-t}$ maps, proving (3).

For each t there are exactly t possible tail positions. Consequently

$$\mathbb{P}(R_n = t) = \frac{t(n-1)_{t-1}}{n^t} = \frac{(n)_t\, t}{n^{t+1}} = \mathbb{P}(C_n = t). \tag{4}$$

which proves the finite distributional identity. It also proves that μ conditional on $R_n = t$ is uniform on $\{0, \ldots, t-1\}$.

No collision during the first m transitions has probability

$$\mathbb{P}(R_n > m) = \frac{(n-1)_m}{n^m} = \prod_{j=1}^{m} \left(1 - \frac{j}{n}\right). \tag{5}$$

For $m = \lfloor x\sqrt{n} \rfloor$, logarithmic expansion gives $\log \mathbb{P}(R_n > m) = -m(m+1)/(2n) + O(m^3/n^2) \to -x^2/2$. The bound $\log(1-z) \le -z$ supplies tail tightness, so (5) converges to the Rayleigh survival function. Equation (4) transfers the limit to C_n. The boundary cells $\mu = 0$, $\lambda = 1$, and $t = n$ are already included in (3), without limiting conventions.

References

[1] B. Harris, *Probability distributions related to random mappings*, Ann. Math. Statist. **31** (1960), 1045–1062, doi:10.1214/aoms/1177705677.

[2] P. Flajolet and A. M. Odlyzko, *Random mapping statistics*, Advances in Cryptology—EUROCRYPT '89, LNCS 434 (1990), 329–354, doi:10.1007/3-540-46885-4_34.